

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250276846

Kind Code

A1

Publication Date

September 04, 2025

Inventor(s)

Whitehead; Stephen et al.

NESTABLE TRASH RECEPTACLE WITH WHEELS

Abstract

The invention involves a wheeled trash receptacle having wheels slidable on an axle between an inboard position where the trash receptacle may be nested inside of a like constructed trash receptacle and an outboard position where the wheels are spaced further apart to provide for stability of the trash receptacle while being moved on its wheels. The wheel spacer is constructed as a one piece member that can be slid over a side surface of the axle to be positioned on either side of either wheel on the trash receptacle. The construction of the wheel spacer also makes removal of the wheel spacer an easy task should the user want to change the configuration of the trash receptacle at any time.

Inventors: Whitehead; Stephen (Elgin, IL), Kopp; Robert (Elburn, IL), Anderson; Torrence (Overland Park, KS)

Applicant: Suncast Technologies, LLC (Palm Beach Gardens, FL)

Family ID: 96880934

Appl. No.: 18/592501

Filed: February 29, 2024

Publication Classification

Int. Cl.: B65F1/14 (20060101)

U.S. Cl.:

CPC B65F1/1473 (20130101); B65F2220/12 (20130101)

Background/Summary

FIELD OF INVENTION

[0001] The present invention generally relates to trash receptacles and, more particularly, to a wheeled trash receptacle having an adjustable wheel position to allow for nesting of one wheeled trash receptacle inside of another wheeled trash receptacle.

BACKGROUND INFORMATION

[0002] Wheeled trash receptacles are known in the art. They generally assist the user in moving a large and heavy amount of trash or refuse by functioning like a two wheeled cart when loaded with weight. This construction allows the user to tip the can toward the user to place the weight over the wheels. The loaded trash receptacle can then be moved to a desired location before being allowed to fall back onto the bottom surface of the trash can, which may lift the wheels away from the ground surface. Unfortunately, wheeled trash receptacles that include widely spaced wheels for stability are typically not nestable and thus shipment adds significant cost to the consumer product. Wheeled trash receptacles that are nestable typically require the wheels to be moved inward, making the trash receptacle unstable when the wheels are used.

[0003] Therefore, current wheeled trash receptacle designs are typically sold without the wheels mounted to the trash can to allow the empty cans to be nested for shipment. Thus, the user must finish assembly of the trash can once purchased and transferred home, requiring tools and small part assembly to complete the trash receptacle. Once assembled, the construction prevents the trash cans from nesting for storage. Thus, the trash s must be stored in a side by side arrangement, consuming a large amount of space.

[0004] U.S. Patent Publication No. 2004/0108666 to Rubbermaid discloses a nestable trash can. The trash can includes an elongated slot in which the axle and wheel arrangement is slidably mounted. In this device, the axle and wheels are slid to a rearward position where the wheels can be moved inwardly toward the center of the trash can. In this position, the trash cans can be partially nested. Use of the trash can requires the axle to be moved to a forward position and wheels moved outwardly. This construction creates several problems, one being the elongated slot in the bottom portion of the trash can. Since the axle extends completely across the trash can there is no way to seal the bottom portion of the can. The elongated slot provides a wide open area for animals, such as rats and mice, to enter the trash can, as well as allow fluids to leak from the trash can.

[0005] U.S. Pat. No. 11,873,161, also to Rubbermaid, discloses a waste receptacle having wheels that can be moved inwardly and outwardly. The wheels are retained in a respective position with a spacer assembly. The spacer assembly includes two pieces, one having spring lock tabs and the other including a socket for the spring tabs. In use, the axle is moved manually to the inward or the outward position. Once in position, the two pieces of the spacer assembly are positioned to interlock with one another by inserting the spring clips into the sockets. Once inserted and locked together the components cannot be disassembled without breaking the components of the spacer.

[0006] Finally, there are ergonomic needs that a wheel spacer must satisfy in order to achieve acceptance by the end user. The system must be easily and quickly assembled using minimal hardware and requiring a minimal number of tools. Further, the system should not require excessive strength to assemble or include heavy component parts. Moreover, the system must assemble together in such a way so as not to detract from the aesthetic appearance of the assembled trash receptacle. Still yet, the wheel spacer should be easily disassembled to allow the user to reconfigure his trash receptacle as needed.

[0007] Thus, the present invention provides a wheeled trash receptacle that includes a one piece wheel spacer which overcomes the disadvantages of prior art. The wheel spacer attachment structure of the present invention not only provides for relative ease in the assembly, it also permits disassembly and repositioning of the wheels on the trash receptacle as desired.

SUMMARY OF THE INVENTION

[0008] Briefly, the invention involves a wheeled trash receptacle having wheels slidable on an axle

between an inboard position where the trash receptacle may be nested inside of a like constructed trash receptacle and an outboard position where the wheels are spaced further apart to provide for stability of the trash receptacle while being moved on its wheels. The wheel spacer is constructed as a one piece member that can be slid over a side surface of the axle to be positioned on either side of either wheel on the trash receptacle. The construction of the wheel spacer also makes removal of the wheel spacer an easy task should the user want to change the configuration of the trash receptacle at any time.

[0009] Accordingly, it is an objective of the present invention to provide a wheeled and nestable trash receptacle.

[0010] It is a further objective of the present invention to provide a wheeled and nestable trash receptacle that includes wheels movable between a nestable and a non-nestable position.

[0011] It is yet a further objective of the present invention to provide a wheel spacer that can be removed and replaced to change the wheel position between the nestable and the non-nestable position.

[0012] It is another objective of the present invention to provide a wheel spacer that snaps over a side surface of the wheel axle.

[0013] It is still yet another objective of the present invention to provide a wheel spacer that is constructed as a single member having a ramping surface to allow the wheel spacer to flex when snapped over the side surface of the wheel axle.

[0014] Other objectives and advantages of this invention will become apparent from the following description taken in conjunction with the accompanying drawings wherein are set forth, by way of illustration and example, certain embodiments of this invention. The drawings constitute a part of this specification and include exemplary embodiments of the present invention and illustrate various objects and features thereof.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0015] FIG. 1 is a top, front, right perspective view of one embodiment of a wheeled trash receptacle utilizing a one piece wheel spacer;

[0016] FIG. 2 is a top, rear, right perspective view of the wheeled trash receptacle utilizing a one piece wheel spacer;

[0017] FIG. 3 is a bottom, rear, right view of the wheeled trash receptacle utilizing a one piece wheel spacer;

[0018] FIG. 4 is a rear, bottom, left view of the wheeled trash receptacle utilizing a one piece wheel spacer;

[0019] FIG. 5 rear is a view of the wheeled trash receptacle utilizing a one piece wheel spacer illustrating the wheels in an outboard position;

[0020] FIG. 6 is a right side view of the wheeled trash receptacle utilizing a one piece wheel spacer illustrating the wheels in an outboard position;

[0021] FIG. 7 is a bottom view of the wheeled trash receptacle utilizing a one piece wheel spacer illustrating the wheels in an outboard position;

[0022] FIG. 8 is a top, rear, right view of the wheeled trash receptacle utilizing a one piece wheel spacer illustrating the wheels in an inboard position;

[0023] FIG. 9 is top, rear, left view of the wheeled trash receptacle utilizing a one piece wheel spacer illustrating the wheels in an inboard position;

[0024] FIG. 10 is a rear view of the wheeled trash receptacle utilizing a one piece wheel spacer illustrating the wheels in an inboard position;

[0025] FIG. 11 is a bottom, rear, left perspective view of the wheeled trash receptacle utilizing a

one piece wheel spacer illustrating the wheels in an inboard position;
[0026] FIG. **12** is a bottom view of the wheeled trash receptacle utilizing a one piece wheel spacer illustrating the wheels in an inboard position;
[0027] FIG. **13** is a top, rear, right perspective view of the axle assembly for the wheeled trash receptacle utilizing a one piece wheel spacer illustrating the wheels in an inboard position on the axle;
[0028] FIG. **14** is a top, front, left view of the axle assembly for the wheeled trash receptacle utilizing a one piece wheel spacer illustrating the wheels in an inboard position on the axle;
[0029] FIG. **15** is a rear view of the axle assembly for the wheeled trash receptacle utilizing a one piece wheel spacer illustrating the wheels in an inboard position on the axle;
[0030] FIG. **16** is a front view of the axle assembly for the wheeled trash receptacle utilizing a one piece wheel spacer illustrating the wheels in an inboard position on the axle;
[0031] FIG. **17** is a top, front, left perspective view of the one piece wheel spacer;
[0032] FIG. **18** is a bottom, rear, right perspective view of the one piece wheel spacer;
[0033] FIG. **19** is a top, front, right perspective view of the one piece wheel spacer;
[0034] FIG. **20** is a bottom, rear, left perspective view of the one piece wheel spacer;
[0035] FIG. **21** is a rear view of the one piece wheel spacer;
[0036] FIG. **22** is a right side view of the one piece wheel spacer;
[0037] FIG. **23** is a front view of the one piece wheel spacer;
[0038] FIG. **24** is a top view of the one piece wheel spacer;
[0039] FIG. **25** is a bottom view of the one piece wheel spacer;
[0040] FIG. **26** is a left side view of the one piece wheel spacer;
[0041] FIG. **27** is a top, front, right perspective view illustrating the trash receptacles in a nested arrangement;
[0042] FIG. **28** is a top, rear, right perspective view illustrating the trash receptacles in a nested arrangement;
[0043] FIG. **29** is a front view illustrating the trash receptacles in a nested arrangement;
[0044] FIG. **30** is a section view taken along lines **30-30** of FIG. **29**;
[0045] FIG. **31** is a side view illustrating the trash receptacles in a nested arrangement;
[0046] FIG. **32** is a section view taken along lines **32-32** of FIG. **31**;
[0047] FIG. **33** is a front, bottom, right perspective view illustrating the trash receptacles in a nested arrangement; and
[0048] FIG. **34** is a rear, bottom, right perspective view illustrating the trash receptacles in a nested arrangement.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0049] While the invention present is susceptible of embodiment in various forms, there is shown in the drawings and will hereinafter be described a presently preferred embodiment with the understanding that the present disclosure is to be considered an exemplification of the invention and is not intended to limit the invention to the specific embodiments illustrated.

[0050] Referring generally to FIGS. **1-34**, a wheeled trash receptacle **100** including a receptacle **10** for receiving and holding trash is disclosed, the receptacle **10** including a bottom wall **12** for supporting the receptacle in an upright position when in contact with a ground surface. A sidewall **14** is integrally formed to and extending completely around a perimeter of the bottom wall **12**; the sidewall **14** including a height **18** suitable for receiving and holding trash. The sidewall **14** terminates in a rim **36** which extends around the perimeter of the sidewall, generally turning outward and then downward to round off or square off the top surface of the sidewall. One or more side handles **38** may be formed or secured to the sidewall **14** for moving and maneuvering the trash receptacle. In at least some embodiments, one or more of the side handles **38** include a cinch clamp **40** that is constructed and arranged to cooperate with a trash bag (not shown) to hold the trash bag taught when positioned inside of the trash receptacle to receive trash. A main handle **16** is provided

at a rear portion of the receptacle **10** for tipping the trash receptacle **100** from a stationary position having the bottom wall **12** against a ground surface to a mobile position wherein the bottom wall is not in contact with the ground surface and a pair of wheels **42** are in contact with the ground surface. The pair of wheels **42** is secured to the receptacle **10** for rotation about an axle **44**. The wheels are preferably positioned in a wheel well **48** so as to not contact the ground surface when the receptacle **10** is in the upright position. The wheels are constructed and arranged to have at least two positions, a first wheel position allowing one wheeled trash receptacle to be nested inside of another like constructed trash receptacle, (see FIGS. **8-16** and **27-34**), and a second wheel position providing greater stability to the receptacle in the mobile position but not allowing nesting (see FIGS. **1-7**). The wheels are configurable by a consumer to be in either the first wheel position or the second wheel position. The wheel position is also changeable at any time by the consumer should they need the trash receptacles to nest for storage or add stability for top heavy loads. In order to provide the changeable wheel position, the trash receptacle is provided with a pair of one piece wheel spacers **20**, wherein one wheel spacer **20** is associated with each respective wheel **42** of the pair of wheels to position the respective wheel in the first wheel position or the second wheel position. In a preferred embodiment, each one piece wheel spacer **20** is constructed and arranged to snap over a side surface **46** of the axle **44** or remove the one piece wheel spacer **20** from the axle **44** by hand. Thus, the consumer can change the position of the wheel spacer **20**, and thus the wheel, by repositioning the one piece wheel spacer **20** on either side of a respective wheel **42**. As noted above, the removal and placement of the one piece wheel spacer **20** can be completed by hand and not require tools that the consumer may not have access to.

[0051] Still referring generally to all of the figures, and more specifically to FIGS. **13-26**, the one piece wheel spacer **20** is illustrated. The one piece wheel spacer **20** is preferably C-shaped, when viewed from an end, having a channel **22** for engaging the side surface **46** of the axle **44**. The channel **22** includes a ramping surface **24** for causing resilient deformation of the one piece wheel spacer **20** as it is slipped over the cylindrical side surface **46** of the axle **44**. In a most preferred embodiment, both sides of the channel **22** include a ramping surface **24** to assist in flexing the one piece spacer **20**. The channel **22** preferably includes a mouth portion **26**, the mouth portion being narrower than a diameter of the axle **44** to resist removal of the one piece spacer **20** from the axle **44**. In at least one embodiment, the one piece spacer **20** is constructed to include a core sleeve **28** and a pair of flanges **30**. In a most preferred embodiment, at least one of the flanges **30** includes rub rails **34** for reducing friction with a respective wheel **42**. The rub rails **34** are preferably elongated and raised projections that rub against a wheel surface as the wheel rotates. This construction reduces friction when rolling a loaded or unloaded trash receptacle by reducing the contact area between the two components. In addition, the polymer or plastic resin used for construction of the one piece wheel spacer **20** may be selected for lower friction when rubbing against the wheel material. Some embodiments include at least one rib **32** that is integrally formed to and extends between the flanges **30** at each end of the one piece spacer **20** to provide structural rigidity to the one piece spacer **20** while maintaining a reduced weight when compared to a solid construction.

[0052] It is to be understood that while a certain form of the invention is illustrated, it is not to be limited to the specific form or arrangement herein described and shown. It will be apparent to those skilled in the art that various changes may be made without departing from the scope of the invention, and the invention is not to be considered limited to what is shown and described in the specification and any drawings/figures included herein.

[0053] One skilled in the art will readily appreciate that the present invention is well adapted to carry out the objectives and obtain the ends and advantages mentioned, as well as those inherent therein. The embodiments, methods, procedures and techniques described herein are presently representative of the preferred embodiments, are intended to be exemplary, and are not intended as limitations on the scope. Changes therein and other uses will occur to those skilled in the art which

are encompassed within the spirit of the invention and are defined by the scope of the appended claims. Although the invention has been described in connection with specific preferred embodiments, it should be understood that the invention as claimed should not be unduly limited to such specific embodiments. Indeed, various modifications of the described modes for carrying out the invention, which are obvious to those skilled in the art, are intended to be within the scope of the following claims.

Claims

1. A wheeled trash receptacle comprising: a receptacle for receiving and holding trash, the receptacle including a bottom wall for supporting the receptacle in an upright position when in contact with a ground surface, a sidewall integrally formed to and extending completely around a perimeter of the bottom wall, the sidewall including a height suitable for receiving and holding trash, a pair of wheels secured to the receptacle for rotation about an axle, the wheels positioned so as to not contact the ground surface when the receptacle is in the upright position, a handle for tipping the receptacle to a mobile position wherein the bottom wall is not in contact with a ground surface and the wheels are in contact with the ground surface, the wheels having at least two positions, a first wheel position allowing one wheeled trash receptacle to be nested inside of another like constructed trash receptacle, a second wheel position providing greater stability to the receptacle in the mobile position but not allowing nesting, the wheels configurable by a consumer in either the first wheel position or the second wheel position.
 2. The wheeled trash receptacle of claim 1 including a one piece wheel spacer associated with each respective wheel of the pair of wheels to position the respective wheel in the first wheel position or the second wheel position.
 3. The wheeled trash receptacle of claim 2 wherein each one piece wheel spacer is constructed and arranged to snap over a side surface of the axle.
 4. The wheeled trash receptacle of claim 3 wherein each one piece wheel spacer is resiliently deformable to snap over the side surface of the axle.
 5. The wheeled trash receptacle of claim 4 wherein each one piece wheel spacer is C-shaped having a channel, the channel including a ramping surface for causing resilient deformation of the one piece wheel spacer as it is slipped over the side surface of the axle.
 6. The wheeled trash receptacle of claim 5 wherein each side of the channel opening includes a ramping surface.
 7. The wheeled trash receptacle of claim 6 wherein the channel includes a mouth portion, the mouth portion being narrower than a diameter of the axle to resist removal of the one piece spacer from the axle.
 8. The wheeled trash receptacle of claim 5 wherein the one piece spacer includes a core sleeve and a pair of flanges.
 9. The wheeled trash receptacle of claim 8 wherein at least one of the flanges includes rub rails for reducing friction with a respective wheel.
 10. The wheeled trash receptacle of claim 9 wherein the rub rails are elongated and raised projections that rub against a wheel surface as the wheel rotates.
 11. The wheeled trash receptacle of claim 10 wherein at least one rib **32** is integrally formed to and extends between the flanges **30** at each end of the one piece spacer to provide structural rigidity to the one piece spacer.
 12. The wheeled trash receptacle of claim 1 wherein the one piece spacer is constructed from polymer.
 13. The wheeled trash receptacle of claim 1 wherein the polymer is plastic resin.
-